

Enumerative Combinatoric Algorithms, summerterm 2025

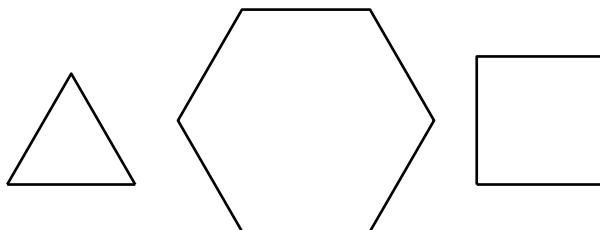
1st Assignment

Surname: Albu Firstname: Sebastian Matr.No.: _____

Remark: Please note that it is mandatory to write the exercise on your own, legibly, and in logical order. The deadline for this exercise is **Wednesday, 2025-03-26 18:00**. Your solution can be handed in physically during the lecture, or via the teach center in form of a pdf-file before the deadline. Please observe also the remarks in the course as well as on the homepage.

This assignment contributes 10% to the course, each task 5%. Provide all details of your arguments in a comprehensive way.

1. **Pigeonhole principle:** Consider (1) an equilateral triangle with side length 1, (2) a hexagon with side length 1, and (3) a square with side length 1.



Compute (integer) values for X_T , X_H , and X_S so that if you (1) place X_T points in the triangle, there are at least two points of distance $\leq \frac{1}{3}$, (2) place X_H points in the Hexagon, there are at least two points of distance $\leq \frac{1}{2}$, (3) place X_S points in the square, there are at least two points of distance $\leq \frac{\sqrt{2}}{3}$. To place a point in a geometric shape means placing it inside the shape or on its boundary. Which lower bounds can you give? Give all details of your argumentation.

Solution. The strategy is to cover each shape with a collection of closed regions (each having diameter not exceeding the given threshold). By the pigeonhole principle, if we place one more point than the number of these regions, at least one region must contain two points whose mutual distance does not exceed its diameter.

1. Equilateral Triangle (Threshold $\frac{1}{3}$)

Divide each side of the equilateral triangle into 3 equal segments and draw lines parallel to the sides. This subdivides the triangle into 9 smaller congruent equilateral triangles, each of side length $\frac{1}{3}$. In any equilateral triangle, the greatest distance between two points is its side length. Therefore, any two points in one of these smaller triangles are at most $\frac{1}{3}$ apart.

Thus, if we place

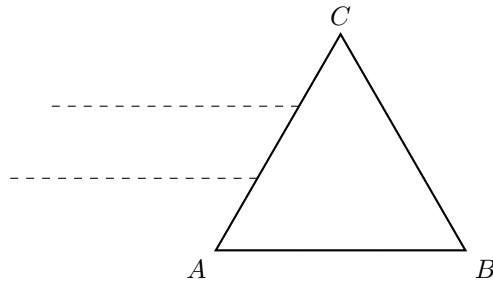
$$9 + 1 = 10$$

points in the triangle, the pigeonhole principle guarantees that at least one of the 9 small triangles contains at least two points with distance at most $\frac{1}{3}$. We set

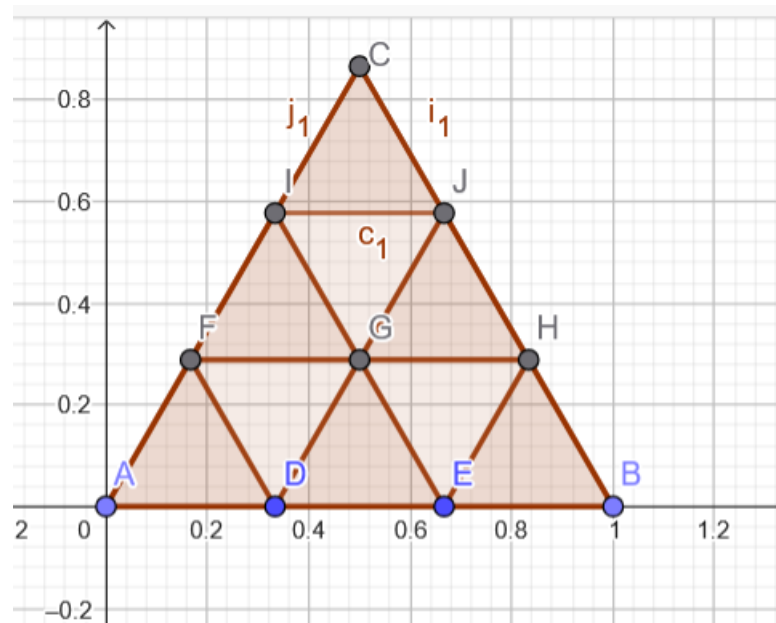
$$X_T = 10.$$

Construction of 9 sub-triangles. Label the vertices of the equilateral triangle ABC , each side of length 1. Divide each side into 3 equal segments of length $\frac{1}{3}$, and draw lines parallel to the sides

through these division points. This yields $3 \times 3 = 9$ small congruent equilateral triangles, each of side $\frac{1}{3}$.



The triangles then look as in the visualization done in GeoGebra below. (Tikz was not cooperating here sadly)

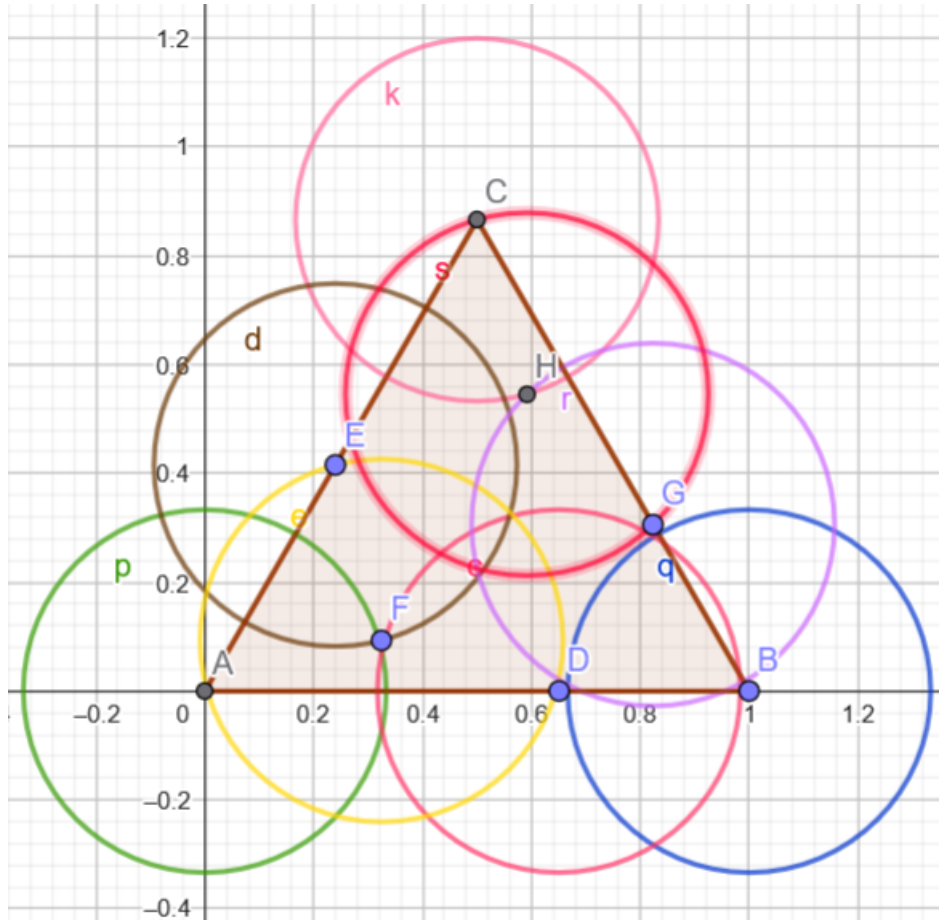


Diameter check. Each small sub-triangle has side length $\frac{1}{3}$, so any two points within it are at distance $\leq \frac{1}{3}$.

Now we aim to find a lower bound for this example.

Claim. There is a way to place 8 points in the unit equilateral triangle so that every pair of points has distance $> \frac{1}{3}$. This shows that we cannot force a close pair with only 8 points, hence $X_T > 8$.

Approximate Coordinates. Below is one explicit arrangement (found by trial and small adjustments).



Hence, 8 points need not force a pair $\leq \frac{1}{3}$. Therefore X_T must exceed 8. Combined with the covering argument $X_T \leq 10$, we see $8 < X_T \leq 10$.

2. Square (Threshold $\frac{\sqrt{2}}{3}$)

Partition the unit square by dividing each side into 3 equal parts, forming a 3×3 grid of 9 smaller squares. Each small square has side length $\frac{1}{3}$ and the maximum distance (diagonal) within each small square is

$$\sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \frac{\sqrt{2}}{3}.$$

Thus, if we place $9 + 1 = 10$ points in the square, the pigeonhole principle guarantees that at least one of the 9 sub-squares contains two points. Since the distance between any two points in a small square is at most $\frac{\sqrt{2}}{3}$, we conclude that with 10 points, there is always a pair with distance

$$\leq \frac{\sqrt{2}}{3}.$$

Hence, we have

$$X_S \leq 10.$$

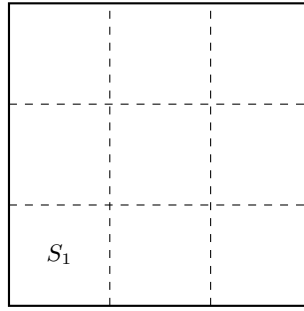


Figure 1: A 3×3 grid subdivision of the unit square.

Explicit 9-Point Configuration (Lower Bound)

We now exhibit an explicit configuration of 9 points in the unit square such that every pair is at distance $> \frac{\sqrt{2}}{3}$. This demonstrates that with only 9 points one may avoid having any pair of points within distance $\frac{\sqrt{2}}{3}$. Hence, X_S must be exactly 10.

For example, consider the following configuration:

$$\begin{aligned} P_1 &= (0, 0), & P_2 &= (1, 0), & P_3 &= (1, 1), & P_4 &= (0, 1), \\ P_5 &= (0.5, 0), & P_6 &= (1, 0.5), & P_7 &= (0.5, 1), & P_8 &= (0, 0.5), \\ P_9 &= (0.5, 0.5). \end{aligned}$$

In this configuration:

- The distance between a vertex (say $P_1 = (0, 0)$) and the midpoint of the adjacent side (say $P_5 = (0.5, 0)$) is 0.5.
- The distance between the center $P_9 = (0.5, 0.5)$ and any edge-midpoint is 0.5.

Since $0.5 > \frac{\sqrt{2}}{3} \approx 0.4714$, every pair of points in this configuration is more than $\frac{\sqrt{2}}{3}$ apart.

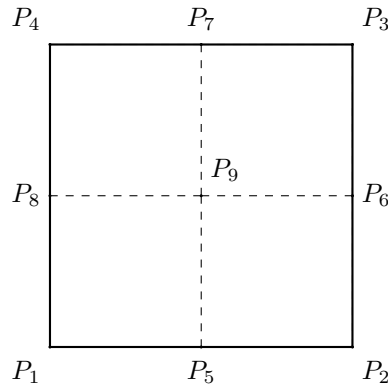


Figure 2: A 9-point configuration in the unit square with all mutual distances $> \frac{\sqrt{2}}{3}$.

Conclusion

The covering argument shows that if 10 points are placed in the square then some two must be at distance

$$\leq \frac{\sqrt{2}}{3}.$$

On the other hand, the explicit 9-point configuration demonstrates that it is possible to have 9 points with every pair at distance

$$> \frac{\sqrt{2}}{3}.$$

Thus, the optimal forcing number is exactly

$$X_S = 10.$$

Summary: - If you have 10 points in the square, by the pigeonhole principle (using the 9 small squares), some pair is at distance $\leq \frac{\sqrt{2}}{3}$. - However, there exists a 9-point configuration with all distances $> \frac{\sqrt{2}}{3}$. Therefore, $X_S = 10$.

3. Regular Hexagon (Threshold $\frac{1}{2}$)

A regular hexagon of side length 1 can be partitioned into 6 large equilateral triangles (each of side length 1). Each of these triangles can further be subdivided by joining the midpoints of its sides, thus splitting it into 4 smaller congruent equilateral triangles with side length $\frac{1}{2}$. Overall, the hexagon is divided into

$$6 \times 4 = 24$$

small equilateral triangles.

Since in each such triangle the maximum distance between any two points is its side length, any two points inside one small triangle are at most $\frac{1}{2}$ apart.

Hence, placing

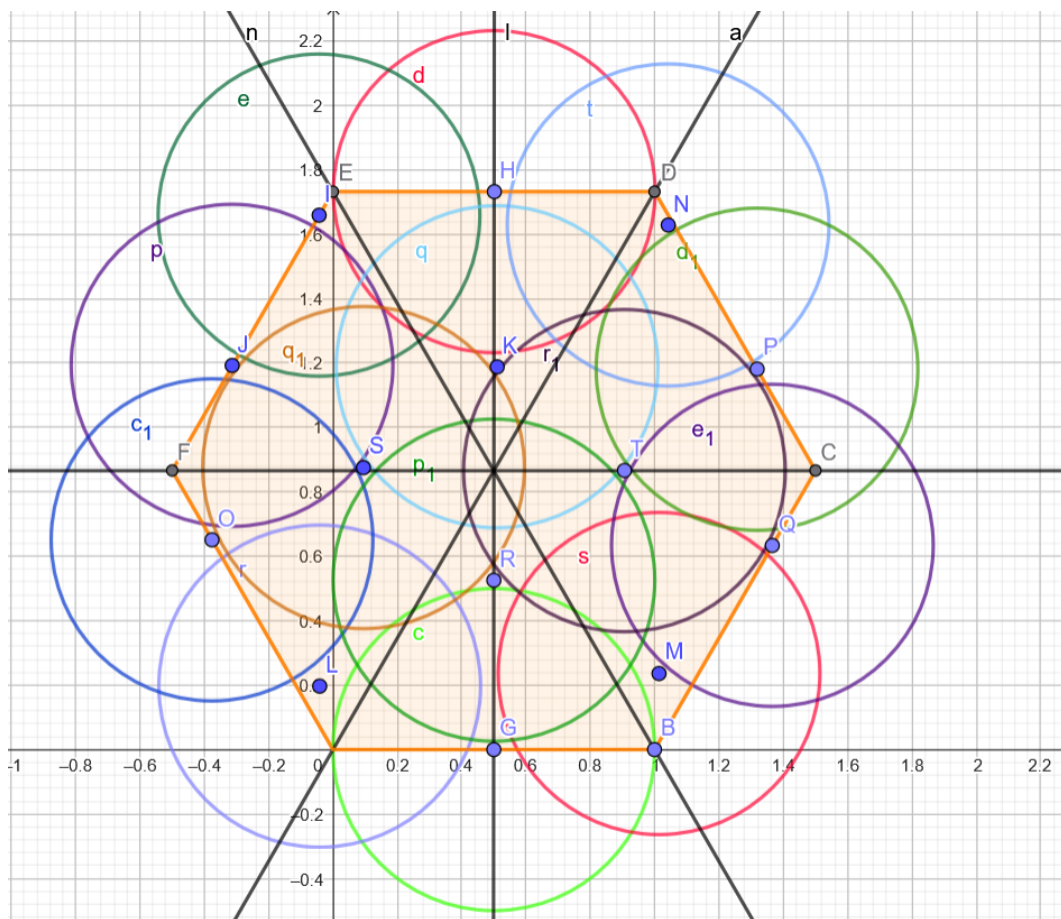
$$24 + 1 = 25$$

points in the hexagon guarantees by the pigeonhole principle that at least one of the 24 regions contains two points within distance $\frac{1}{2}$. We therefore set

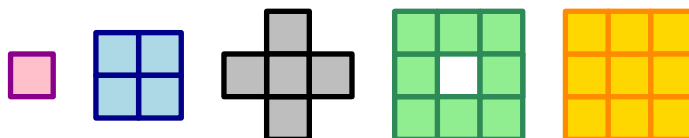
$$X_H = 25.$$

We can, in theory, play around with geometries and also find another tighter bound. However, that would require not using the pigeonhole principle. We now aim to prove a lower bound.

We now present an example of 16 points arranged inside the regular hexagon (side 1) such that every pair of points is at a distance $> \frac{1}{2}$. Each of those 16 points with a circle of radius $> \frac{1}{2}$ contains exactly one point, meaning that $X_H > 16$. (Note some inaccuracies due to manually working with the points in GeoGebra, but this design is possible, leading to $16 < X_H \leq 25$).



2. **Polyominoes:** By *simple polyominoes* we denote polyominoes where there is no difference between the fixed and the free version. The following picture shows simple polyominoes for $n = 1, 4, 5, 8$, and 9.



For which general n do simple polyominoes exist? Argue your answers in detail.

Simple Polyominoes

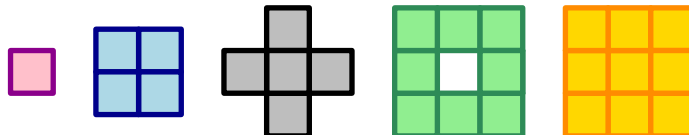
A *polyomino of size n* is a connected shape made up of n unit squares (cells) on the integer grid, joined edge-to-edge. When counting *free* polyominoes, we identify two shapes if one can be transformed into the other by a combination of translations, rotations (by 90° increments), and reflections. By contrast, when counting *fixed* polyominoes, we only identify shapes if one can be translated onto the other (no rotations or reflections allowed).

Definition (Simple Polyomino). A polyomino is called *simple* if its free and fixed versions coincide. Equivalently, *every* rotation or reflection of that polyomino can be aligned with the original shape by a *pure translation* on the grid. Hence, under the full dihedral group of symmetries (the 8 isometries generated by rotating the square by 90° and reflecting), the polyomino remains “the same” up to a shift in the plane.

Examples. The problem statement shows examples of simple polyominoes for $n = 1, 4, 5, 8$, and 9 :

- $n = 1$: a single square.
- $n = 4$: the 2×2 block (it is unchanged by rotation/reflection).
- $n = 5$: a “plus” shape (centered cross).
- $n = 8$: the 3×3 square with its center cell removed (an 8-cell ring).
- $n = 9$: the 3×3 block.

All of these remain congruent to themselves (up to translation) under any rotation or reflection. You can see all of them below.



Claim and Construction

Claim. Simple n -polyominoes exist **if and only if** $n \equiv 0 \pmod{4}$ or $n \equiv 1 \pmod{4}$. In other words, there is a simple polyomino exactly for those n that are multiples of 4 or one more than a multiple of 4.

Two Infinite Families. We exhibit two natural infinite families that show existence for all $n \equiv 0 \pmod{4}$ and $n \equiv 1 \pmod{4}$:

1. **“Plus” shapes for $n = 1 + 4k$.**

Consider the polyomino shaped like a centered cross with one central cell and equally long arms in the four cardinal directions. Concretely, for a nonnegative integer k :

$$\text{plus shape with arm length } k \implies (1 + 4k)\text{-cell polyomino.}$$

For example:

$$k = 0 \Rightarrow n = 1, \quad k = 1 \Rightarrow n = 5, \quad k = 2 \Rightarrow n = 9, \quad \dots$$

This “plus” is invariant (up to translation) under any 90° rotation or reflection about the center.

2. **“Ring” shapes for $n = 4k$.**

Consider the $(m \times m)$ -square “frame” (ring) obtained by removing the $(m - 2) \times (m - 2)$ interior, where $m \geq 2$. Its total number of cells is

$$m^2 - (m - 2)^2 = m^2 - (m^2 - 4m + 4) = 4m - 4 = 4(m - 1).$$

By letting $k = m - 1$, we get a ring of size $4k$. For instance:

$$m = 2 \Rightarrow k = 1, n = 4 \quad (\text{the } 2 \times 2 \text{ block}), m = 3 \Rightarrow k = 2, n = 8 \quad (\text{the } 3 \times 3 \text{ ring}), m = 4 \Rightarrow k = 3, n = 12, \quad \dots$$

Such a ring is also symmetric under all rotations by 90° and reflections (the center of the ring is the center of symmetry), and hence is simple.

Thus, for every integer $k \geq 0$, we obtain:

$$\text{a simple polyomino of size } n = 4k \quad \text{and} \quad \text{a simple polyomino of size } n = 1 + 4k.$$

These cover all $n \equiv 0 \pmod{4}$ and $n \equiv 1 \pmod{4}$.

Why No Simple Polyomino for $n \equiv 2$ or $3 \pmod{4}$?

Intuitively, being *simple* forces the polyomino to match itself under a full 90° rotation (and reflections) up to a pure shift. In particular:

- A 90° rotation preserves the shape *only* if the bounding box is a square and the shape is arranged so that, upon rotation, it can slide (translate) exactly onto its original position. One finds that such “4-fold” symmetry enforces a kind of concentric structure (like the ring) or a centered cross structure (like the plus).

- Counting the squares layer by layer from the center shows that either there is a single center cell (giving $1 + 4k$ squares) or there is a hollow center (giving $4k$ squares). In neither scenario can we get a total of $n \equiv 2 \pmod{4}$ or $n \equiv 3 \pmod{4}$.

Hence no shape of size 2, 3, 6 ($\equiv 2 \pmod{4}$), 7 ($\equiv 3 \pmod{4}$), 10 ($\equiv 2 \pmod{4}$), 11 ($\equiv 3 \pmod{4}$), etc. can be invariant under all dihedral symmetries up to translation. Here it helps to consider the polyomino of size 2 and 3, where we can trivially see that a rotation of 90 degrees already provides a proof by contradiction in case we assume these polyominoes to be simple.

Conclusion

Simple polyominoes exist exactly for those integers n that are multiples of 4 or one more than a multiple of 4.

Two explicit infinite families are:

Ring shapes (size $4k$), and Plus shapes (size $1 + 4k$).

(Also, obviously, we exclude the case of $n = 0$, since that polyomino would make no sense).

Enumerative Combinatoric Algorithms SS 2025

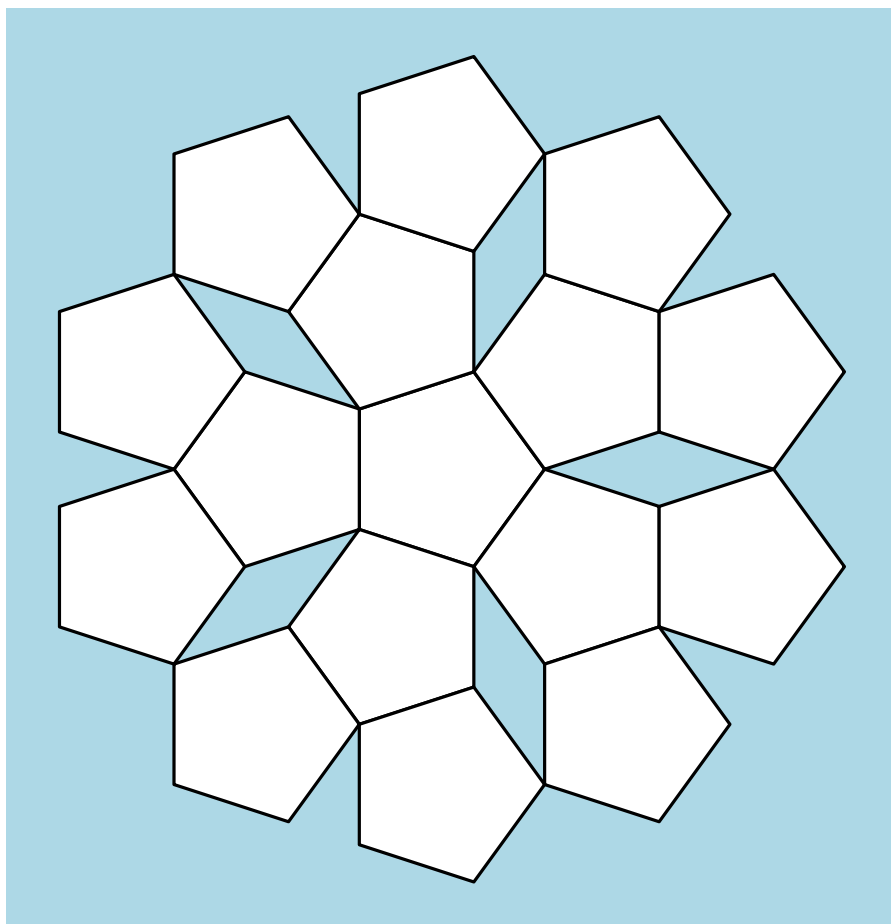
2nd Assignment

Surname: Albu Firstname: Sebastian Matr.No.: _____

Remark: Please note that it is mandatory to write the exercise on your own, legibly and in logical order. The deadline for this exercise is **Wednesday, 2025-05-07 18:00**. Your solution can be handed in physically during the lecture or via the teach center in form of a pdf-file before the deadline. Please observe also the remarks in the course as well as on the homepage.

This assignment contributes 10% to the course. Provide all details of your arguments in a comprehensive way.

Pólya-Redfield Enumeration Theorem: Consider the following penrose-style pattern which consists of 16 pentagons:



In how many different ways can you color the 16 cells with k colors? Define the operations for “different” using all meaningful rotations and reflections. Give an explicit formula for all $k \geq 2$, and compute the concrete values for $k = 1, 2, 3, 4$. Give all intermediate steps of your considerations and reasoning in a clean and easy to follow way.

1 Pólya–Redfield on a Penrose-Style 16-Pentagon Patch using Burnside’s Lemma

We have a patch of 16 congruent regular pentagons arranged with full dihedral symmetry of order 10 (the symmetry group D_5). Label the cells by

$$X = \{1, 2, \dots, 16\},$$

and let $G = D_5$ act by the rotations

$$r^i: X \rightarrow X \quad (i = 0, 1, \dots, 4)$$

and the reflections

$$s r^i: X \rightarrow X \quad (i = 0, 1, \dots, 4).$$

We wish to count the number of orbits of colorings $X \rightarrow \{1, \dots, k\}$ under this action.

The Dihedral Group D_5

The *dihedral group* D_n is the full symmetry group of a regular n -gon, consisting of n rotations and n reflections. Concretely,

$$D_n = \langle r, s \mid r^n = e, s^2 = e, srs = r^{-1} \rangle,$$

where

- e is identity by $e = r^0$ (or r^x with $x \equiv 0 \pmod{n}$).
- r is rotation by $\frac{2\pi}{n}$,
- s is a reflection across some fixed axis.

Thus $|D_n| = 2n$. In our case $n = 5$, so

$$D_5 = \{e, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\},$$

and $|D_5| = 10$.

Burnside’s Lemma

Burnside’s Lemma gives the number of orbits of a finite set X under the action of a finite group G . In full generality:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|,$$

where

$$\text{Fix}(g) = \{x \in X : g \cdot x = x\}$$

is the set of points fixed by g . The factor $1/|G|$ is simply the averaging over all group elements. In our example, $|G| = 10$, so we divide by 10.

1. Cycle-Index Computation

By inspecting the action on the 16 cells (one center pentagon, five in the first ring, ten in the outer ring) one finds:

- **Identity** $e = r^0$: fixes all 16 cells, so $|\text{Fix}(e)| = k^{16}$.
- **Rotation** r^i , $i = 1, 2, 3, 4$: the center is fixed (1-cycle), the 5 in the middle ring form one 5-cycle, and the 10 in the outer ring split into two 5-cycles. Hence $|\text{Fix}(r^i)| = k^{1+1+2} = k^4$, $i = 1, \dots, 4$.
- **Reflections** $s r^i$, $i = 0, \dots, 4$: each fixes the center (1 cycle), fixes exactly one of the 5 in the inner ring. Since 5 is odd, each reflection axis passes through the center cell and one inner cell (fixing those two), pairing the remaining 4 in the inner ring into two 2-cycles, and pairing the 10 outer cells into five 2-cycles. Hence $|\text{Fix}(s r^i)| = k^{1+1+2+5} = k^9$, $i = 0, \dots, 4$.

By Burnside’s Lemma,

$$\{\text{inequivalent colorings}\} = \frac{1}{G} \sum_{g \in G} \text{Fix}(g) = \frac{1}{10} (k^{16} + 4k^4 + 5k^9).$$

2. Explicit Formula and Values

$$N(k) = \frac{k^{16} + 4k^4 + 5k^9}{10}.$$

In particular:

$$N(1) = \frac{1 + 4 + 5}{10} = 1,$$

$$N(2) = \frac{2^{16} + 4 \cdot 2^4 + 5 \cdot 2^9}{10} = \frac{65536 + 64 + 2560}{10} = 6816,$$

$$N(3) = \frac{3^{16} + 4 \cdot 3^4 + 5 \cdot 3^9}{10} = \frac{43\,046\,721 + 324 + 98\,415}{10} = 4\,314\,546,$$

$$N(4) = \frac{4^{16} + 4 \cdot 4^4 + 5 \cdot 4^9}{10} = \frac{4\,294\,967\,296 + 1024 + 1\,310\,720}{10} = 429\,628\,904.$$

Enumerative Combinatoric Algorithms

3rd Assignment

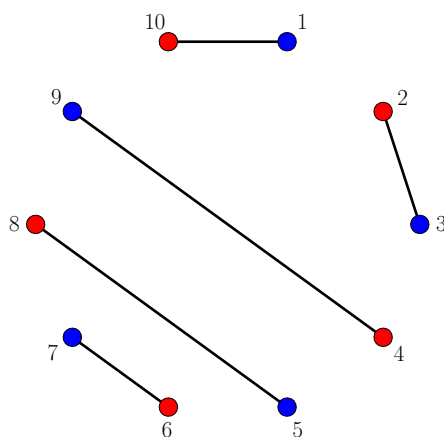
summer term 2025

Surname: Sebastian Firstname: Albu Matr.No.: _____

Remark: Please note that it is mandatory to write the exercise on your own, legibly, and in logical order. The deadline for this exercise is **Wednesday, 2025-06-04 18:00**. Your solution can be handed in physically during the lecture, or via the teach center in form of a pdf-file before the deadline. Please observe also the remarks in the course as well as on the homepage.

This assignment contributes 10% to the course. Provide all details of your arguments in a comprehensive way.

Let S be a set of n points, n even, in the plane in convex position. The points are labeled 1 to n in clockwise order and they are colored red if their label is even and blue if their label is odd.



Let $\mathcal{M}$ be the set of all crossing free perfect matchings which can be drawn on S such that every edge is bicolored. An edge is bicolored if it connects points of different colors, that is, there is no edge connecting two red (or two blue) points. Describe how Reverse Search can be used to enumerate all elements of $\mathcal{M}$, that is, all crossing free perfect matchings of S with bicolored edges.

Remark: Provide all the details of your reasoning. Prove the correctness of all your statements.

1. Root Matching and Fingerprint

Definition 1 (Root Matching). *For even n , let*

$$M_0 \in \mathcal{M} \quad M_0 = \{(1, 2), (3, 4), \dots, (n-1, n)\}.$$

All edges are consecutive, hence crossing-free and bicolored.

Definition 2 (Fingerprint of a Matching). *Given $M \in \mathcal{M}$, sort its edges as*

$$e_1 \prec e_2 \prec \dots \prec e_{n/2},$$

where $(a, b) \prec (c, d)$ iff $a < c$ or $a = c \wedge b < d$. We call the sequence $(e_1, \dots, e_{n/2})$ the fingerprint of M .

2. Neighborhood via Quadrilateral Flips

Definition 3 (General Flip). *Let M be a perfect, non-crossing, bicolored matching. Suppose $(a, b), (c, d) \in M$ with $a < b < c < d$ around the hull, and that $\{(a, d), (b, c)\}$ is also crossing-free and bicolored. Then*

$$M' = (M \setminus \{(a, b), (c, d)\}) \cup \{(a, d), (b, c)\}$$

is called a flip of M on the quadrilateral (a, b, c, d) .

Define

$$\Gamma(M) = \{M' \mid M' \text{ obtained by one quadrilateral-flip of } M\}.$$

Invariants of a Flip

- $M' = M = n/2$ (perfectness).
- Crossing-free, since the four points form a convex quadrilateral.
- Bicolored, as points alternate parity around the hull.
- Symmetric: flipping twice returns to the original matching.

By classical results on non-crossing matchings, the graph $(\mathcal{M}, \Gamma)$ is connected.

3. Predecessor via Fingerprint Reduction

Definition 4 (Predecessor Function f): Let

$$\text{fp}(M) = (e_1, \dots, e_{n/2})$$

be the fingerprint of M , and let

$$\text{fp}(M_0) = (r_1, \dots, r_{n/2}), \quad r_i = (2i-1, 2i).$$

Find the smallest index k such that $e_k \neq r_k$. Since $r_k \prec e_k$, there exists a flip replacing e_k by r_k . Perform that flip to obtain M' , and define $f(M) = M'$.

Lemma 1. *For all $M \neq M_0$, $f(M) \in \Gamma(M)$ and $\text{fp}(f(M)) \prec \text{fp}(M)$. Hence the directed edges $(M, f(M))$ form an acyclic spanning tree rooted at M_0 .*

Proof: By construction the first mismatching entry in the fingerprint strictly decreases in lex order, so no cycles occur and M eventually reaches M_0 .

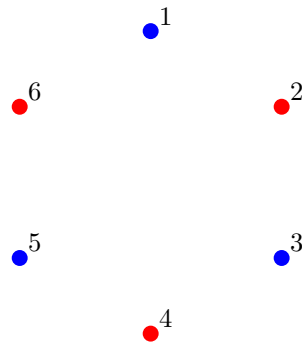
4. Algorithm (Sketch)

ReverseSearch(M) : $\begin{cases} \text{output } M, \\ \text{for each } N \in \Gamma(M) \text{ in order of the new edge : } \quad \text{if } f(N) = M \text{ then ReverseSearch}(N). \end{cases}$

Call ReverseSearch(M_0) to enumerate all matchings exactly once.

5. Worked Example for $n = 6$

We now consider this example with 6 points (not with the 10 points as on the first page, as that one would have been the same procedure but just longer). Place points $1, \dots, 6$ around a circle (odd=blue, even=red):

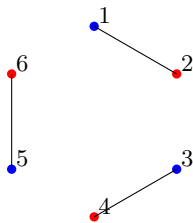


The five matchings ($C_3 = 5$) are

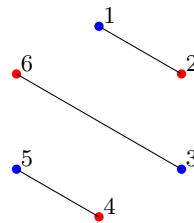
$$\begin{aligned} M_0 &= \{(1, 2), (3, 4), (5, 6)\}, \\ M_1 &= \{(1, 2), (3, 6), (4, 5)\}, \\ M_2 &= \{(1, 4), (2, 3), (5, 6)\}, \\ M_3 &= \{(1, 6), (2, 3), (4, 5)\}, \\ M_4 &= \{(1, 6), (2, 5), (3, 4)\}. \end{aligned}$$

Their fingerprints:

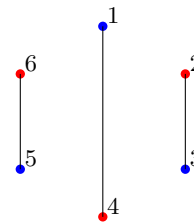
$$\begin{aligned} \text{fp}(M_0) &= (12, 34, 56), \\ \text{fp}(M_1) &= (12, 36, 45), \\ \text{fp}(M_2) &= (14, 23, 56), \\ \text{fp}(M_3) &= (16, 23, 45), \\ \text{fp}(M_4) &= (16, 25, 34). \end{aligned}$$



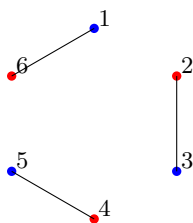
$$M_0 = \{(1, 2), (3, 4), (5, 6)\}$$



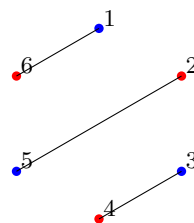
$$M_1 = \{(1, 2), (3, 6), (4, 5)\}$$



$$M_2 = \{(1, 4), (2, 3), (5, 6)\}$$



$$M_3 = \{(1, 6), (2, 3), (4, 5)\}$$



$$M_4 = \{(1, 6), (2, 5), (3, 4)\}$$

Traversal:

- From M_0 (root), flips on runs $(1, 2, 3, 4)$ and $(3, 4, 5, 6)$ yield

$$M_2 \text{ (new edge 14)}, \quad M_1 \text{ (new edge 36)}.$$

Since $14 \prec 36$, visit M_2 first, then backtrack and visit M_1 .

•

- At M_2 ($\text{fp} = (14, 23, 56)$), the first candidate quad is $(1, 2, 3, 4)$. Flipping it

$$(1, 4), (2, 3) \rightarrow (1, 2), (3, 4)$$

yields M_0 , but since $f(M_0) \neq M_2$, it is not a child. Next we try quad $(3, 4, 5, 6)$; that flip

$$(3, 4), (5, 6) \rightarrow (3, 6), (4, 5)$$

produces M_3 , and since $f(M_3) = M_2$, it is the (only) child of M_2 .

- At M_3 ($\text{fp} = (16, 23, 45)$), mismatch at index 2. Flip on $(3, 4, 5, 6)$: $(3, 4), (5, 6) \rightarrow (3, 6), (4, 5)$ gives M_4 .
- M_4 has no new child (all flips return to visited matchings).
- Backtrack: return through $M_3 \rightarrow M_2$ then finally visit M_1 .

Thus the DFS order

$$M_0 \rightarrow M_2 \rightarrow M_3 \rightarrow M_4 \text{ [backtrack]} \rightarrow M_1$$

visits all five matchings exactly once.